

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

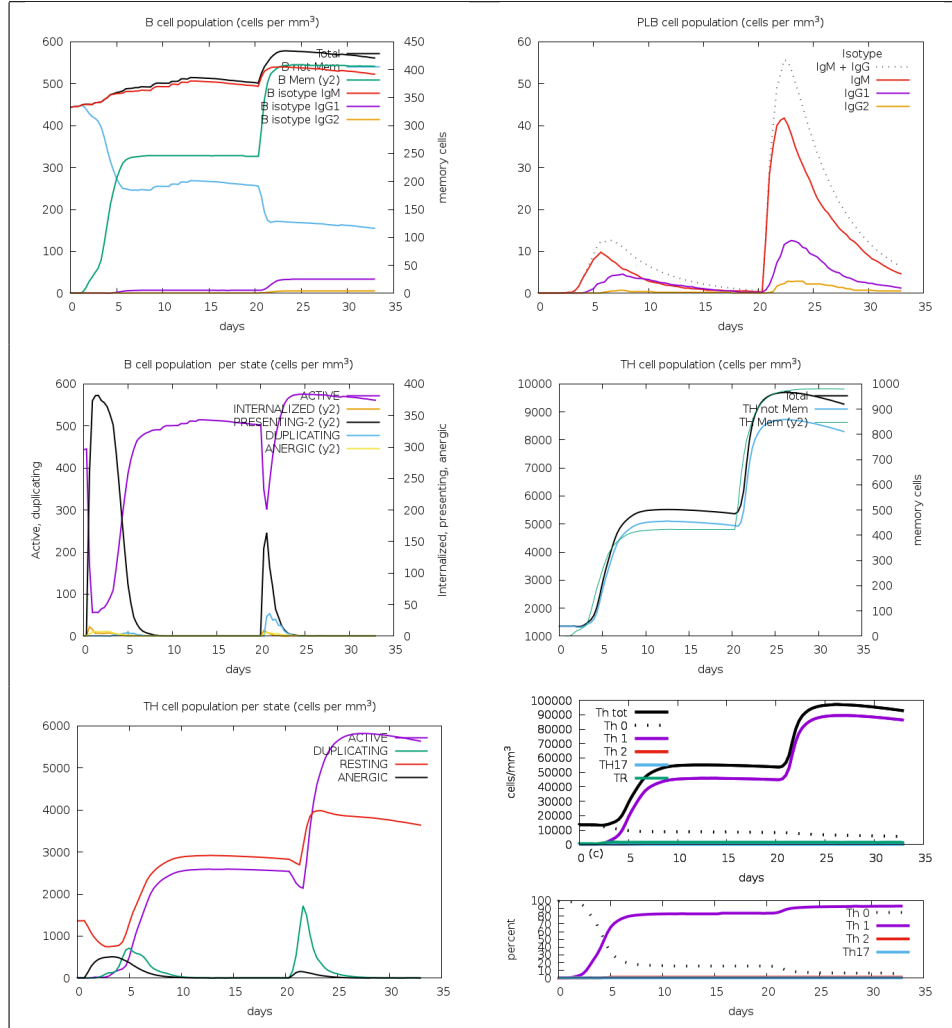


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

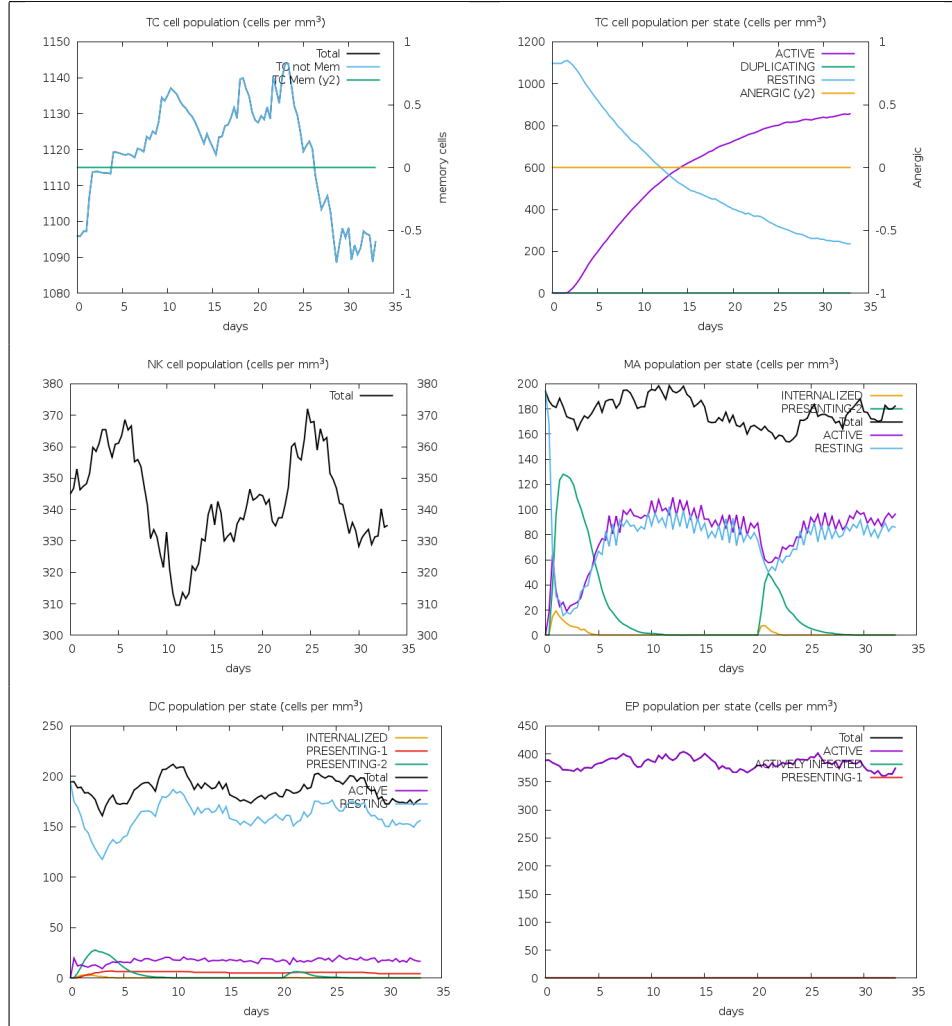


Figure 2: Legend: symbols as figure above.

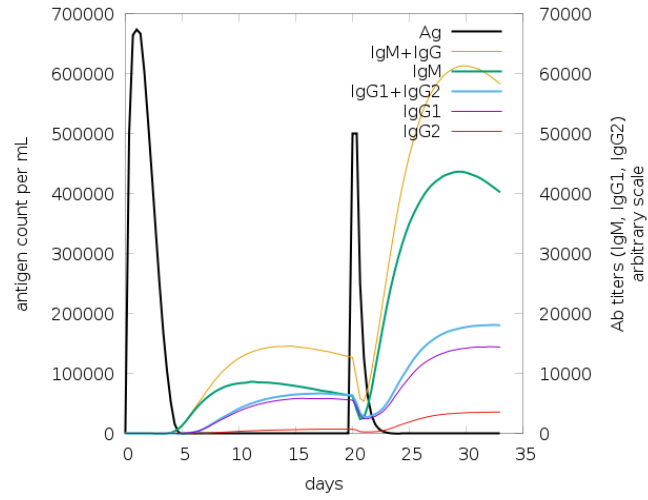


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

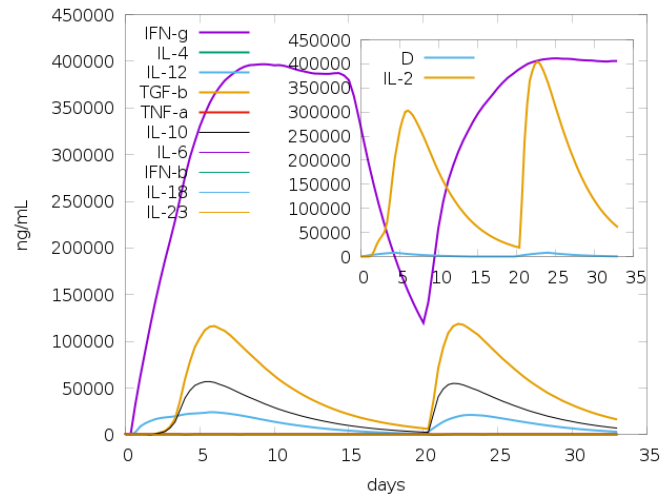


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Parker_B_scale_epitope_find
Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

0000000000001111110000000110111000000000011111100000000000000001110000000000
00000000000100
0000000000001000
00000000000

Epitopes of protein 0 -----
1] pos=12 len=7 AGGSDQA
2] pos=42 len=7 HASGEGP

DoPeptideList_I:
Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 4 MHCI molecules

Read class I peptide list from file? NO

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 0 -----
0] pos= 14 score=0.052226 unnormalised=5.7896000000 GSDQATRVY
1] pos= 136 score=0.002044 unnormalised=0.2266000000 AAAAAIALY
2] pos= -1 score=0.945730 unnormalised=104.8410000000 non-binding event

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPGIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 0 -----

0]	pos=	14	score=0.052226	unnormalised=5.7896000000	GSDQATRVY
1]	pos=	136	score=0.002044	unnormalised=0.2266000000	AAAAAIALY
2]	pos=	-1	score=0.945730	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPGIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 0 -----

0]	pos=	27	score=0.012497	unnormalised=1.5682000000	GPSQAPPGI
1]	pos=	51	score=0.036028	unnormalised=4.5212000000	NPISKTA AAA
2]	pos=	72	score=0.019844	unnormalised=2.4902000000	PPRGPPLSA
3]	pos=	75	score=0.009341	unnormalised=1.1722000000	GPPLSAAAA
4]	pos=	103	score=0.025924	unnormalised=3.2532000000	VPFVAAAAA
5]	pos=	117	score=0.032402	unnormalised=4.0662000000	APQLFAAAA
6]	pos=	135	score=0.008592	unnormalised=1.0782000000	YAAAAAIAL
7]	pos=	205	score=0.019923	unnormalised=2.5002000000	IPNLFAAAA
8]	pos=	-1	score=0.835449	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPGIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 0 -----

0]	pos=	27	score=0.012497	unnormalised=1.5682000000	GPSQAPPGI
1]	pos=	51	score=0.036028	unnormalised=4.5212000000	NPISKTA AAA
2]	pos=	72	score=0.019844	unnormalised=2.4902000000	PPRGPPLSA
3]	pos=	75	score=0.009341	unnormalised=1.1722000000	GPPLSAAAA

4]	pos= 103	score=0.025924	unnormalised=3.2532000000	VPFVAAAAA
5]	pos= 117	score=0.032402	unnormalised=4.0662000000	APQLFAAAA
6]	pos= 135	score=0.008592	unnormalised=1.0782000000	YAAAAAIAL
7]	pos= 205	score=0.019923	unnormalised=2.5002000000	IPNLFAAAA
8]	pos= -1	score=0.835449	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

=====

Allele: DRB1_0101
Pseudo sequence: KAFHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGGSDQATRVYQDLGGPSQAPPIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPLS
AAAAAGTYKRVTEKAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAAASCFTASVSTVVSATAAAAAATPDIL
PVVTRNLNAI

Epitopes of protein 0 -----

0]	pos= 2	score=0.012370	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001687	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 22	score=0.011821	unnormalised=3.2345600000	YQDLGGPSQ
3]	pos= 53	score=0.019712	unnormalised=5.3935600000	ISKTA AAAA
4]	pos= 78	score=0.025724	unnormalised=7.0385600000	LSAAAAAGT
5]	pos= 87	score=0.007651	unnormalised=2.0935600000	YKRVTEKAA
6]	pos= 105	score=0.029400	unnormalised=8.0445600000	FVAAAAAYQ
7]	pos= 106	score=0.023465	unnormalised=6.4205600000	VAAAAAYQA
8]	pos= 112	score=0.030968	unnormalised=8.4735600000	YQAYAAPQL
9]	pos= 120	score=0.031103	unnormalised=8.5105600000	LFAAAAAAT
10]	pos= 121	score=0.045298	unnormalised=12.3945600000	FAAAAAATV
11]	pos= 129	score=0.036998	unnormalised=10.1235600000	VKGMQSYAA
12]	pos= 132	score=0.017347	unnormalised=4.7465600000	MQSYAAAAA
13]	pos= 135	score=0.044293	unnormalised=12.1195600000	YAAAAAIAL
14]	pos= 143	score=0.015915	unnormalised=4.3545600000	LYLQQNWAA
15]	pos= 145	score=0.022193	unnormalised=6.0725600000	LQQNWAAAA
16]	pos= 149	score=0.042831	unnormalised=11.7195600000	WAAAAANST
17]	pos= 162	score=0.002056	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 178	score=0.015688	unnormalised=4.2925600000	FNSILFALL
19]	pos= 182	score=0.037203	unnormalised=10.1795600000	LFALLAAAA
20]	pos= 183	score=0.054936	unnormalised=15.0315600000	FALLAAAAA
21]	pos= 185	score=0.034963	unnormalised=9.5665600000	LLAAAAAAA
22]	pos= 186	score=0.032912	unnormalised=9.0055600000	LAAAAA AAA
23]	pos= 196	score=0.028626	unnormalised=7.8325600000	FKSLSSSTEF
24]	pos= 205	score=0.000784	unnormalised=0.2145600000	IPNLFAAAA
25]	pos= 208	score=0.030533	unnormalised=8.3545600000	LFAAAAAAS
26]	pos= 209	score=0.024437	unnormalised=6.6865600000	FAAAAAASC
27]	pos= 218	score=0.027924	unnormalised=7.6405600000	FTASVSTVV
28]	pos= 222	score=0.006665	unnormalised=1.8235600000	VSTVVSATA
29]	pos= 225	score=0.020709	unnormalised=5.6665600000	VVSATAAAA

30]	pos= 226	score=0.021963	unnormalised=6.0095600000	VSATAAAAA
31]	pos= -1	score=0.241823	unnormalised=66.1680000000	non-binding event

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Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013617_5_hGsdzHtDOEOM.FSA_1_001

AKFVAAWTLKAAAGSDQATRVYQDLGGPSQAPPGIDELAQQHASGEGPGINPISKTA AAAALERGPPGPRRPPRGPPLS
AAAAAGTYKRVTEKAAAAADPYQVPFVAAAAAYQAYAAPQLFAAAAAATVKGMQSYAAAAAIALYLQQNWAAAAANSTNI
TAVVRAQGLDAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAASCFTASVSTVVSATAAAAAATPDIL
PVTNRNLNAI

Epitopes of protein 0 -----

0]	pos= 2	score=0.012370	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001687	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 22	score=0.011821	unnormalised=3.2345600000	YQDLGGPSQ
3]	pos= 53	score=0.019712	unnormalised=5.3935600000	ISKTA AAAA
4]	pos= 78	score=0.025724	unnormalised=7.0385600000	LSAAAAAGT
5]	pos= 87	score=0.007651	unnormalised=2.0935600000	YKRVTEKAA
6]	pos= 105	score=0.029400	unnormalised=8.0445600000	FVAAAAAYQ
7]	pos= 106	score=0.023465	unnormalised=6.4205600000	VAAAAAYQA
8]	pos= 112	score=0.030968	unnormalised=8.4735600000	YQAYAAPQL
9]	pos= 120	score=0.031103	unnormalised=8.5105600000	LFAAAAAAT
10]	pos= 121	score=0.045298	unnormalised=12.3945600000	FAAAAAATV
11]	pos= 129	score=0.036998	unnormalised=10.1235600000	VKGMQSYAA
12]	pos= 132	score=0.017347	unnormalised=4.7465600000	MQSYAAAAA
13]	pos= 135	score=0.044293	unnormalised=12.1195600000	YAAAAAIAL
14]	pos= 143	score=0.015915	unnormalised=4.3545600000	LYLQQNWAA
15]	pos= 145	score=0.022193	unnormalised=6.0725600000	LQQNWAAAA
16]	pos= 149	score=0.042831	unnormalised=11.7195600000	WAAAAANST
17]	pos= 162	score=0.002056	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 178	score=0.015688	unnormalised=4.2925600000	FNSILFALL
19]	pos= 182	score=0.037203	unnormalised=10.1795600000	LFALLAAAA
20]	pos= 183	score=0.054936	unnormalised=15.0315600000	FALLAAAAA
21]	pos= 185	score=0.034963	unnormalised=9.5665600000	LLAAAAAAA
22]	pos= 186	score=0.032912	unnormalised=9.0055600000	LAAAAAAA
23]	pos= 196	score=0.028626	unnormalised=7.8325600000	FKSLSTEF
24]	pos= 205	score=0.000784	unnormalised=0.2145600000	IPNLFAAAA
25]	pos= 208	score=0.030533	unnormalised=8.3545600000	LFAAAAAAS
26]	pos= 209	score=0.024437	unnormalised=6.6865600000	FAAAAAASC
27]	pos= 218	score=0.027924	unnormalised=7.6405600000	FTASVSTVV
28]	pos= 222	score=0.006665	unnormalised=1.8235600000	VSTVVSATA
29]	pos= 225	score=0.020709	unnormalised=5.6665600000	VVSATAAAA
30]	pos= 226	score=0.021963	unnormalised=6.0095600000	VSATAAAAA
31]	pos= -1	score=0.241823	unnormalised=66.1680000000	non-binding event